Human-AI Co-Creation in Visual Communication Design: A Framework and Empirical Study

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Abstract

Generative artificial intelligence is changing visual communication design from a tool-assisted activity into a collaborative process in which designers and AI systems share work across ideation, generation, evaluation, and refinement. Existing studies discuss AI support for image generation, prompting, and design exploration, but they offer limited guidance on how human creativity, design judgment, and AI-generated visual outputs can be organized within a complete design workflow. This paper proposes a human-AI co-creation framework for visual communication design. The framework contains four stages: design problem definition, AI-assisted visual exploration, human-led selection and refinement, and final design evaluation. It also clarifies human-led tasks, AI-supported tasks, and shared review tasks. An empirical study with professional visual designers was conducted to examine how AI participation affects design efficiency, creative exploration, visual quality, and designer experience. The results show that human-AI co-creation supports faster visual prototyping and wider idea generation, while human designers remain necessary for design intention, semantic accuracy, aesthetic judgment, and contextual fit. The findings are presented as trend-level evidence because the study used a modest sample and one poster-design brief, which limits generalizability. The study provides a structured framework and empirical evidence for understanding human-AI collaboration in intelligent visual communication design.

CCS Concepts

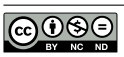
• **Human-centered computing → Empirical studies in interaction design**; **Empirical studies in HCI**; **Command line interfaces**.

Keywords

human-AI co-creation, visual communication design, generative AI, intelligent design, design workflow, empirical study

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1 Introduction

Generative artificial intelligence (AI) is changing how visual communication designers move from a brief to a finished visual artifact. Compared with traditional computer-aided tools, generative AI models can produce visual alternatives, suggest image compositions, and support repetitive production tasks. As noted by Li et al. [6], User Experience (UX) and visual design professionals increasingly view generative AI as more than an operational tool in everyday design practice. The use of AI in visual design also creates practical problems. Designers need to decide when to rely on automation and when to keep direct control over meaning, style, and context. AI can produce visual material quickly, but human review is still needed for semantic accuracy and aesthetic judgment. Prior work has examined collaboration strategies in creative prompting [4]. Chiou et al. [2] also show that co-ideation between designers and image generators can shape the quality of design outcomes. However, the current literature still lacks a structured account of how human and AI roles should be arranged across a full visual communication design process. This gap is important because the value of AI depends not only on image generation, but also on how generated materials are interpreted, selected, edited, and evaluated by designers. To address this issue, this paper proposes a four-stage Human-AI Co-Creation Framework and evaluates it through an empirical study comparing traditional human-only design with a human-AI collaborative workflow. The contribution of this paper is threefold. First, it defines a stage-based framework that separates problem definition, AI-assisted visual exploration, human-led refinement, and final evaluation. It adds role boundaries for human responsibility, AI support, and transfer to related design tasks. Second, it reports an empirical comparison between a traditional group and a co-creation group in a controlled visual design task. The analysis is framed as an initial validation of trend-level differences. Third, it discusses how AI changes the designer's work from manual production toward prompting, selection, editing, and judgment, while preserving the designer's responsibility for meaning and context.

2 Related Work

2.1 Generative AI in Visual and Graphic Design

Generative AI has been used in several areas of visual design. In color design, Hou et al. [5] introduced generative color-concept associations, allowing designers to map abstract semantic concepts to visual color palettes. In marketing and graphic design, Liu et al.

[9] showed how AI-infused exploration can help everyday users create promotional posts. Beyond 2D graphics, generative approaches have also been used for early exploration in spatial and interior design, where users can quickly visualize conceptual layouts [11].

2.2 Human-AI Collaborative Creation

The shift toward co-creation calls for models that describe how designers and AI systems share work. Cao et al. [1] proposed compositional structures for human-AI environments, allowing designers to build complex concepts from modular elements. AI-assisted diagramming and mapping can also support human-centered design processes by helping designers organize causal pathways and logic structures [12]. Lin and Long [8] explored speculative design futures and discussed how AI creativity may change the scope of human imagination.

2.3 Evaluation, Trust, and Cognitive Impacts

Although AI can improve productivity, its cognitive and psychological effects on designers require careful study. Wadinambiarachchi et al. [10] investigated how generative AI influences design fixation and divergent thinking, showing both inspiration and possible overreliance on AI outputs. Gmeiner et al. [3] examined the challenges designers face when learning to co-create with AI-based manufacturing design tools. Li [7] used the UTAUT model to analyze designers' behavioral intentions and found that perceived risk and anxiety affect the adoption of AI-generated content in professional workflows.

3 The Human-AI Co-Creation Framework

Based on the literature above, we propose a four-stage framework for visual communication design. The framework is shown in Figure 1.

(1) **Design Problem Definition:** The human designer establishes the core communication goals, target audience, and brand guidelines. AI can assist by summarizing briefs or generating mood boards based on text descriptions.
(2) **AI-Assisted Visual Exploration:** Designers use text-to-image and image-to-image prompting [4] to rapidly generate a wide array of visual concepts, exploring divergent aesthetic directions [10].
(3) **Human-Led Selection and Refinement:** The designer exercises aesthetic and semantic judgment to select the most viable AI outputs. Using compositional tools [1], the designer edits, remixes, and refines the visuals to ensure they align with the initial problem definition.
(4) **Final Evaluation:** The collaborative output is evaluated against the design criteria. The human designer acts as the final gatekeeper for semantic accuracy, ethics, and overall visual quality.

The framework treats AI as a source of visual alternatives rather than as an independent author. In the first stage, the designer translates the brief into goals, audience information, visual constraints, and evaluation criteria. In the second stage, AI tools expand the range of possible images and layouts. In the third stage, the designer reduces this large set of outputs through selection, correction, and recomposition. In the final stage, the output is checked against

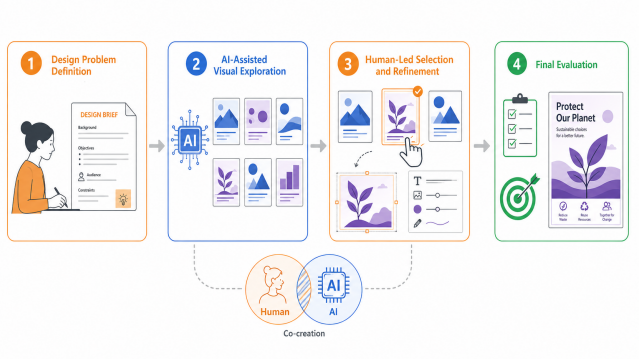


Figure 1: Human-AI co-creation framework for visual communication design.

communication goals, visual quality, and ethical requirements. This order keeps human judgment visible at the beginning and end of the process, where meaning and responsibility are most important. The boundary of the framework is as follows. Humans lead goals, audience meaning, aesthetic judgment, ethics, and final approval. AI supports brief summarization, mood-board search, concept variation, layout trials, and partial image generation. Shared review is needed during selection and refinement, especially when attractive AI outputs drift away from the intended message. The framework can transfer to tasks with clear goals, multiple early concepts, and professional review before release.

4 Empirical Study

4.1 Methodology

To evaluate the proposed framework, an empirical study was conducted with 30 professional visual designers. As shown in Figure 2, participants were randomly divided into two groups: the *Traditional Group* (n=15), who used standard digital design software (e.g., Adobe Illustrator/Photoshop) without generative AI, and the *Co-Creation Group* (n=15), who followed our proposed framework using text-to-image AI tools alongside standard software. Both groups were given an identical brief: design a visual identity and promotional poster for an upcoming technology ethics conference. The task was time-boxed to 120 minutes. Participants were asked to record the time spent in each phase and to save intermediate design outputs. This setup allowed the study to compare not only final quality, but also the amount of exploration and the distribution of effort across the design process.

4.2 Data Collection and Metrics

We measured three primary categories:

- **Design Efficiency:** Time spent (in minutes) across different phases of the design task.
- **Creative Exploration:** The number of distinct visual concepts generated before finalization, and divergent thinking scores assessed by external expert judges (1–10 scale).
- **Subjective Evaluation:** Expert ratings on visual quality, semantic accuracy, and self-reported designer satisfaction (1–7 Likert scale).

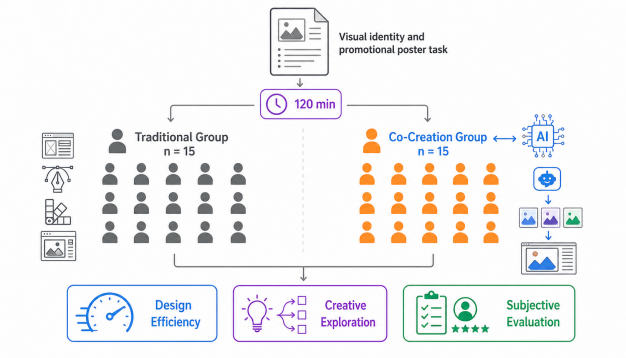


Figure 2: Empirical study design, including participant grouping, task setting, time limit, and evaluation metrics.

Table 1: Design Efficiency: Average Time Spent per Phase (Minutes)

Design Phase	Traditional Group	Co-Creation Group
Problem Definition	15.4	16.2
Ideation & Drafting	45.2	18.5
Refinement & Editing	35.8	42.4
Final Output Prep.	18.1	14.3
Total Time	**114.5**	**91.4**

The expert evaluation was conducted after the design session. Judges reviewed the final posters without knowing which group produced each work. They rated aesthetic appeal, semantic accuracy, and output novelty using the same scoring sheet. Participant satisfaction was collected immediately after the task to reduce recall bias. The intermediate design files were used to count the number of distinct concepts and to check whether participants revised, discarded, or reused AI-generated materials during the process.

Given the sample size and single poster scenario, the analysis reports descriptive statistics only, without formal significance tests or variance measures. It compares group means, mean differences, relative changes, and concept expansion to identify trend-level evidence across aesthetic appeal, semantic accuracy, and novelty. This framing ensures that the exploratory observations are not overstated as definitive conclusions.

5 Results

5.1 Design Efficiency

As shown in Table 1, the Co-Creation group required less total time than the Traditional group, with a 20.2% reduction from 114.5 mins to 91.4 mins. Ideation and Drafting decreased from 45.2 mins to 18.5 mins, while Refinement increased. This suggests that AI speeds early exploration but still requires human correction.

5.2 Creative Exploration and Divergence

Table 2 shows that AI increased the range of explored concepts. The Co-Creation Group produced 2.56 times as many concepts as the Traditional Group and received a higher Divergent Thinking Score. This is consistent with prior findings that generative AI can reduce design fixation during early ideation [10].

Table 2: Creative Exploration Metrics

Metric	Traditional Group	Co-Creation Group
Concepts Generated	3.4	8.7
Divergent Thinking (1–10)	6.2	8.4
Concept Discard Rate (%)	25%	78%

Table 3: Expert Ratings and Designer Satisfaction (1–7 Scale)

Evaluation Criteria	Traditional Group	Co-Creation Group
Aesthetic Appeal	5.1	6.4
Semantic Accuracy	6.0	5.8
Output Novelty	4.8	6.5
Designer Satisfaction	5.2	6.1

5.3 Quality and Subjective Evaluation

We assessed the final outcomes using a 7-point Likert scale. As Table 3 shows, the Co-Creation group scored higher in Aesthetic Appeal and Output Novelty. The overall quality score increased from 5.30 to 6.23. However, Semantic Accuracy was slightly lower than in the Traditional group, which supports the need for human-led refinement because AI outputs may drift away from the intended message [7].

Across the main indicators, AI support produced a 23.1 min average time reduction, 5.3 more concepts, a 2.2-point average gain in divergent thinking, and a 0.93-point average gain in overall quality. The only negative direction observed at the trend level was a 0.2-point decrease in semantic accuracy.

Figure 3 provides a simplified cross-indicator summary of the main quantitative trends. To avoid visual overload, the figure highlights high-level trajectories and should be read together with the detailed tables. Time allocation is reported in Table 1, creative exploration in Table 2, and subjective quality in Table 3. Taken together, the tables and figure indicate that AI participation changes the allocation of design labor. It reduces the effort needed to produce initial alternatives and increases the need for selection and refinement. This pattern suggests that productivity gains should not be interpreted as full automation. Instead, they reflect a redistribution of time from initial drafting to critical editing and final judgment.

6 Discussion

The empirical results support the proposed Human-AI Co-Creation Framework. Generative AI was most useful in the AI-Assisted Visual Exploration stage, where designers could test a broader set of visual directions in less time [2]. The high concept discard rate (78%) in the Co-Creation group, shown in Table 2, indicates that designers treated AI outputs as disposable sketches rather than final products. At the same time, the results show why human judgment remains central. The slightly lower semantic accuracy score in the Co-Creation group suggests that AI cannot replace the designer's understanding of communication goals. As highlighted by Cao et al. [1], human designers often need to deconstruct and recompose

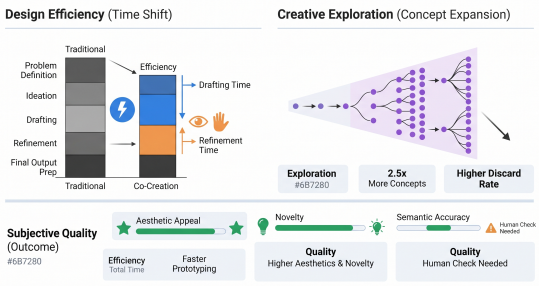


Figure 3: Simplified summary of key empirical trends comparing the Traditional Group and the Co-Creation Group. Indicators are organized by efficiency, exploration, and quality.

AI outputs to meet professional standards. In this sense, the designer's value shifts from manual drafting alone toward prompting, curation, aesthetic judgment, and semantic alignment.

6.1 Mechanism of Creative Expansion and Semantic Drift

The paired result of stronger creative divergence and slightly lower semantic accuracy can be theoretically explained by prompt quality, designer experience, and AI output type. Short prompts often turn complex goals into simple visual cues, resulting in outputs rich in style but weaker in message alignment. Designers with less prompt experience may also tend to keep attractive images that only partly match the brief. However, it is important to note that this mechanism is primarily based on author interpretation. The current study lacks process-level evidence—such as comprehensive prompt histories, systematic revision traces, or stratified analyses by designer profile—to robustly prove these behaviors. Future research must gather these process-level metrics to solidify the precise mechanism behind semantic drift and creative expansion.

For design education and practice, this finding suggests that AI literacy should be taught with visual judgment. Designers need to write prompts and reject attractive but unsuitable results. A useful AI-assisted workflow should include checkpoints for brief interpretation, output selection, correction, and final review.

This study has notable limitations. The experimental scale itself is small, including only 30 designers and a single poster brief, meaning the generalizability of these findings remains limited. As noted, the quantitative differences observed are best treated as trend-level evidence rather than strong statistical conclusions. Different design tasks may place different demands on AI tools and human judgment. Future studies should test more domains and examine designers with different experience levels. Larger studies should also collect prompt histories, process logs, standard deviations, confidence intervals, and inferential test results to validate these initial findings.

7 Conclusion

This paper presented a four-stage Human-AI Co-Creation Framework for visual communication design and evaluated it through an empirical study. The revised framework clarifies human-led judgment, AI-assisted generation, and shared review. The findings show that AI can improve efficiency, broaden exploration, and increase novelty. The results also show that human designers remain necessary for semantic accuracy and contextual relevance. Because the empirical scale is limited, the quantitative outcomes are positioned as trend-level initial validation that require broader testing with significance testing and process-level metrics.

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